

Penny Game

Choose a head or tails and cover it up. Other player has to guess and correctly to win.

Def A game in strategic form:

$S^1, \dots, S^n$ finite set where S^i is the strategy of the i th player

There is a function $H : S^1 \times \dots \times S^n \rightarrow R^n$

Payoff function which takes as input the strategy profile (tuple of strategies chosen by the players)

Objective of each player is to maximize this payoff

Strategic Form of Penny Game

$n = 2$

Let S^1 be prof strategys and $S^1 = \{H, T\}$

Let S^2 be student strategys and $S^2 = \{H, T\}$

S^1/S^2	Head	Tail
Head	$(-1/ + 1)$	$(+1/ - 1)$
Tail	$(+1/ - 1)$	$(-1/ + 1)$

Def A game in extensive form

T tree, edges are labeled “action labels”

Internal verticies are partitioned into n sets and the i th set are the decision nodes of the i th player

Note that a player can play twice if their action moves towards another node in their set

Payoff function $H : \text{leaves} \rightarrow R^n$

Ex tic tac toe

Where the graph is constructed as the edges represent the turns each player could take, and the levels represent whose turn it is. Keeping track of overlal board state and which moves are and aren't possible. Go until a leaf of the tree (which means game over) with a corresponding payoffs for the two players.

Extensive vs Strategic Form

Extensive form may seem like an extension of the strategic form game but actually this isn't the case because in the case of extensive form games players know the previous players choices. In actuality, any game in extensive form can be a game in strategic form due to the "hidden" nature of strategic form

Conversion of Extensive Form to Strategic Form

The big difference is that the strategies are revealed at the same time in Strategic. We can model this as a series of conditionals essentially. Essentially each player's node sets are indexed and each player's strategy is a long tuple of what they would play at each of these nodes. Note that most of these actions would NOT be played but its what WOULD be played IF the game progresses to that node.

Def 2 Person Zero Sum Games Games in strategic form where $n = 2$ and "zero sum" is because $H(s_1, s_2) = R^2 = (h_1, h_2) \rightarrow h_1 = -h_2$

Note that player 1 is trying to maximize h_1 and minimize h_2 thus we are saying the player 1 is the maximizing player and player 2 is the minimizing player

Def A mixed strategy for player i is a probability distribution over S^i

If all players are performing a mixed strategy then we are able to find the expected value of H $E(H)$

let $x(y)$ be a mixed strategies for maximizing (minimizing) player

$$h(x, y) = E_{x,y}(h(i, j)) = \sum_{i \in S^1} \sum_{j \in S^2} x(i)y(j)h(i, j)$$

Suppose we rewrite the penny game payoff function to be of this format (assuming 1 is the maximizing player):

S^1/S^2	Head	Tail
Head	-1	+1
Tail	+1	-1

Def A pure strategy for player i is they will always do strategy $s \in S^i$

Note that this can be represented as a mixed strategy with strategy s having a probability of 1 with all others as 0

Def Game Value = $h(i, j)$ if both players play optimally (or $h(x, y)$)

Suppose the following game:

S^1/S^2	a	b	c
a	3	1	2
b	1	0	1

Regardless of S^2 , player 1 should play a because its always higher than playing b.

For player 2, they should always play b because Regardless of S^1 it results in a h that is less than or equal to any of the other outcomes.

How does prior information affect the game?

How game value affected if max/min player has to reveal ahead of time his pure/mixed strategy?

Revealing Pure Strategy

If max player plays $i \in S^+$ then game value $\min_{j \in S^-} h(i, j)$

→ best choice for max player if they have to give their strategy is:

$$\max_{i \in S^+} \min_{j \in S^-} h(i, j)$$

And this is symmetrical for the min player:

$$\min_{j \in S^-} \max_{i \in S^+} h(i, j)$$

Revealing Mixed Strategy

If max player plays according to mixed strategy x

$$\text{Game value} = \min_{j \in S^-} h(x, j)$$

$$\rightarrow \text{best choice for max player is } \max_x \min_{j \in S^-} h(x, j)$$

And this is symmetrical for the min player:

$$\min_y \max_{i \in S^+} h(i, y)$$

To find an optimal mixed strategy we need an algorithm that takes in h and outputs x (or y)

Algorithm for finding optimal mixed strategy

Suppose we are finding mixed strategy x for max player

Decision var: $x(i), i \in S^+$

$$\max \min_{j \in S^-} \sum_{i \in S^+} x(i)h(i, j)$$

$$\text{s.t. } \sum_{i \in S^+} x(i) = 1$$

$$x(i) \geq 0$$

Which is equivalent to:

$$\max \gamma$$

$$\text{s.t. } \sum_{i \in S^+} h(i, j)x(i) \geq \gamma, (j \in S^-)$$

$$\sum_{i \in S^+} x(i) = 1$$

$$x(i) \geq 0$$

Finding dual of this

$$\max \gamma$$

$$\text{s.t. } \gamma - \sum_{i \in S^+} h(i, j)x(i) \leq 0, (j \in S^-)$$

$$\sum_{i \in S^+} x(i) = 1$$

$$x(i) \geq 0$$

suppose the first constraint can be labelled as $y(j)$ and the second as β

$$\min \beta$$

$$\text{s.t. } -\sum_{j \in S^-} h(i, j)y(j) + \beta \geq 0, (i \in S^+)$$

$$\sum_{j \in S^-} y(j) = 1$$

$$y(j) \geq 0$$

This is the algorithm for the minimizing player

Thus, in the case of revealing mixed strategies, the max player revealing their mixed strategy is the same as the min player revealing their mixed strategy.

This is the von Neuman theorem

Corollary (Yao's principle)

$$\max_x \min_{j \in S^-} h(x, j) = \min_y \max_{i \in S^+} h(i, y)$$

Thus by weak duality:

$$\max_x \min_{j \in S^-} h(x, j) \leq \forall y, \max_{i \in S^+} h(i, y)$$

vonNeuman Theorem

The theorem is simply:

$$\max_x \min_{j \in S^-} h(x, j) = \min_y \max_{i \in S^+} h(i, y)$$

In other words, the max player revealing their mixed strategy is the same as the min player revealing their min strategy at optimal.

Corollary of Yao's principle:

$$\max_x \min_{j \in S^-} h(x, j) \leq \forall y, \max_{i \in S^+} h(i, y)$$

$$\min_y \max_{i \in S^+} h(i, y) \geq \forall x, \min_{j \in S^-} h(x, j)$$

Ski rental

Rent per day a pair of skis or you buy skis for b .

Suppose renting is 1 dollar per day:

$$\min \text{cost} = \min\{b, D\} \text{ where } D \text{ is renting for } D \text{ days.}$$

The skier's strategy is an integer t which is the day when buy the skis ($1 \leq t < \infty$)

$$\text{cost } s_b(D, t) = \{t \leq D \rightarrow t - 1 + b, t > D \rightarrow D\}$$

Game min player for skier: t

Max player weather: D

$$h(D, t) = \frac{s_b(D, t)}{\min\{b, D\}}$$

In otherwords this is how much the skier payed vs paying optimally / minimum amount

Pure strategy for the skier is to fix t so max player would set $D = t$

$$h(t, t) = \frac{t-1+b}{\min\{b, t\}} \geq 1 + 1 - \frac{1}{\min\{b, d\}}$$

Mixed strategy y distribution over t

Q: What is the power of randomness in this game?

By yao's corollary:

$$\min_y \max_D h(D, y) \geq \forall x, \min_t h(x, t)$$

Note that x is a distribution over D

Suppose $x = \Pr[D = d] : 1/2$ for $d = 1$ and $d = 3b$, 0 else.

Intuitively this is $t = 2$ because for $1/2$ i spend 1 and for $1/2$ I spend $1 + b$

Thus our $h(x, t) = 1/2 + 1/2 \frac{1+b}{b} = 1 + \frac{1}{2b}$

Supposedly if we actually find the optimal, $h(x, y) = \frac{e}{e-1}$